

Sensory Processing Sensitivity is associated with lower vagally-mediated heart rate variability

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Methods

One hundred and thirty-eight female undergraduates participated in the study, but 11 were excluded due to missing/poor-quality ECG, or due to identified arrhythmias that negated accurate HRV analysis. The sample was further restricted to young adults, resulting in the exclusion of one participant above 25 years of age. The mean age of this final sample (N = 126) was 19.10 years (SD = 1.48) and 93.7% identified as Hispanic. To assess SPS, participants were administered the Highly Sensitive Person Scale. Heart Rate Variability was based log transformed RMSSD and High Frequency HRV (HfHRV) values obtained from 3-minute ECG recordings (Biopac system) analyzed using Kubios HRV Scientific software.

Discussion

We found the SPS is associated with lower vagally-mediated HRV, even when controlling for age. This association suggests that those with higher sensory processing sensitivity might experience less cognitive adaptability and less parasympathetic activity as indexed by HRV, which may contribute to findings of higher psychological stress and poor physiological outcomes. Our sample was limited to female young adults (18-25) Future research would therefore benefit from examining this SPS-HRV relationship in older samples given that RMSSD and HfHRV have been found to decline rapidly between the ages of 20 and 40. Research has also identified sex differences in RMSSD and HfHRV for young adults, suggesting a need for replication of this study with males. Finally, further exploration of the relationship between high SPS, low HRV, other physiological markers, physical and mental health outcomes is needed to determine points of intervention for those individuals at greater risk of poor health.

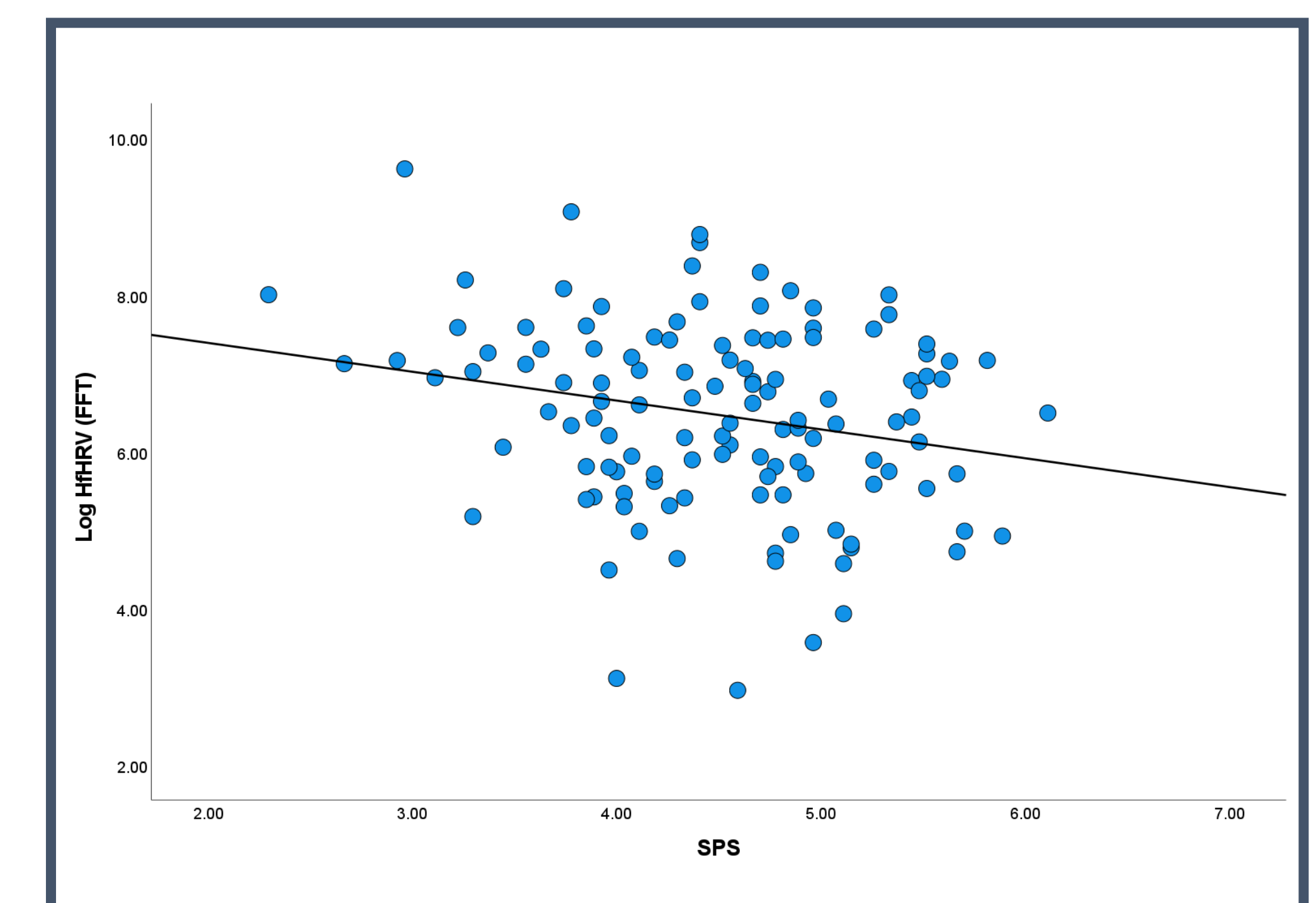
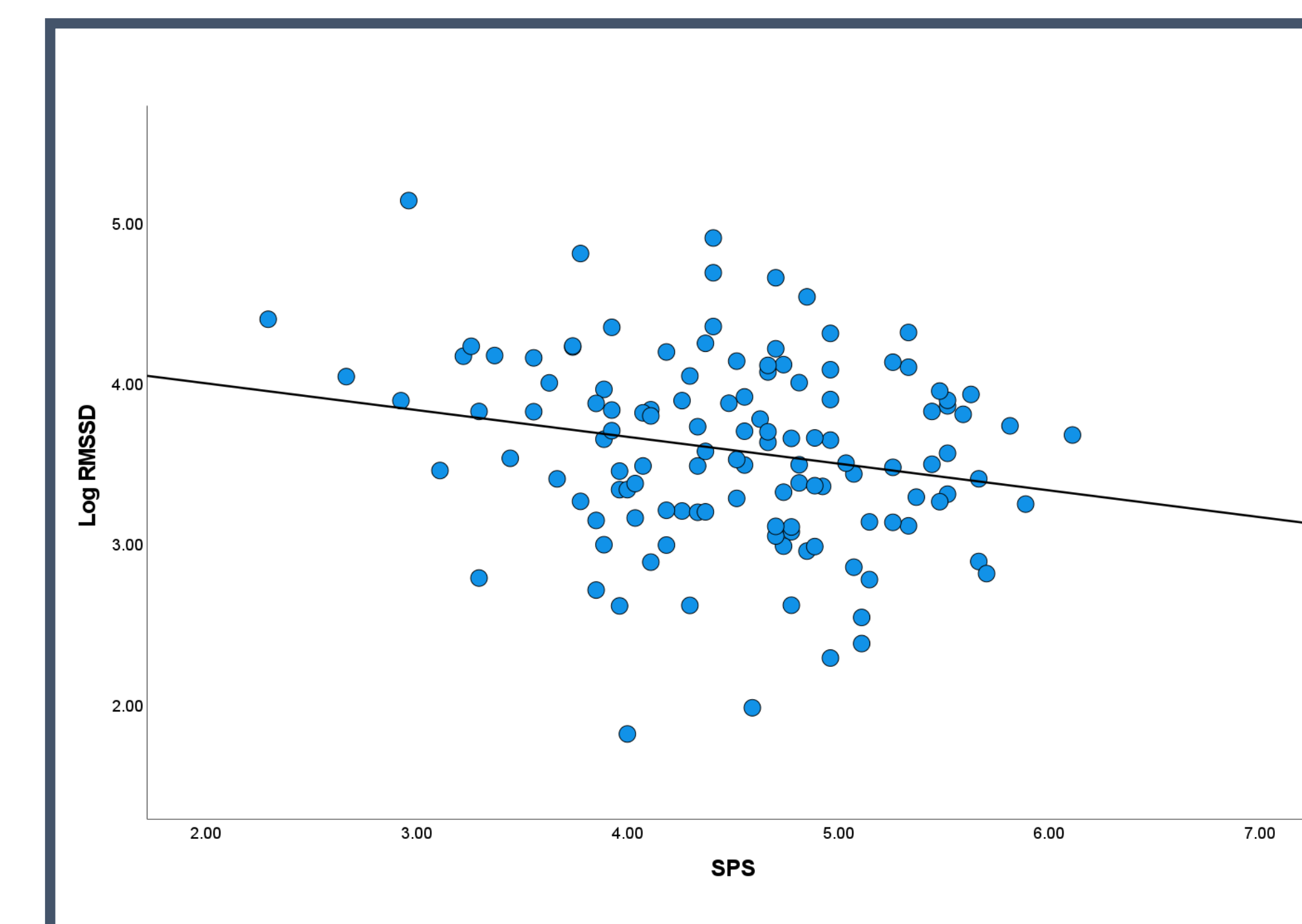
Introduction

Sensory Processing Sensitivity (SPS), an innate trait related to individual differences in stimuli processing¹, has previously been shown to be associated with poorer psychological and poorer physical health^{2,3}.

However, the link between SPS and physical health has largely been based on self-report measures. We therefore investigated the association of SPS with heart rate variability (HRV), a measure of beat-to-beat changes controlled by the brain via the vagus nerve believed to index the brain's ability to adaptively respond to changing environments (neurovisceral integration theory)⁴. As a result, greater HRV is generally viewed as representative of better health. We therefore hypothesized that greater SPS would be associated with lower HRV.

Results

A statistically significant negative relationship was identified between SPS scores and the two indices of HRV (log RMSSD; log HfHRV), even after controlling for age ($r(118) = -.211, p < .05$; $r(118) = -.229, p < .05$, respectively).



References

1. Aron, E. N., & Aron, A. (1997, Aug). Sensory-processing sensitivity and its relation to introversion and emotionality. *Journal of Personality and Social Psychology*, 73(2), 345-368. <https://doi.org/10.1037//0022-3514.73.2.345>
2. Benham, G. (2006, May). The highly sensitive person: Stress and physical symptom reports. *Personality and Individual Differences*, 40(7), 1433-1440. <https://doi.org/10.1016/j.paid.2005.11.021>
3. Greven, C. U., Lionetti, F., Booth, C., Aron, E. N., Fox, E., Schendan, H. E., Pluess, M., Bruining, H., Acevedo, B., Bitttebier, P., & Homberg, J. (2019, Mar). Sensory Processing Sensitivity in the context of Environmental Sensitivity: A critical review and development of research agenda. *Neuroscience and Biobehavioral Review*, 98, 287-305. <https://doi.org/10.1016/j.neubiorev.2019.01.009>
4. Thayer, J. F., & Lane, R. D. (2009). Claude Bernard and the heart-brain connection: Further elaboration of a model of neurovisceral integration. *Neuroscience Biobehavioral Review*, 33, 81-88. [doi:10.1016/j.neubiorev.2008.08.004](https://doi.org/10.1016/j.neubiorev.2008.08.004)
5. Natarajan, A., Pantelopoulos, A., Emir-Farinas, H., & Natarajan, P. (2020). Heart rate variability with photoplethysmography in 8 million individuals: A cross-sectional study. *The Lancet Digital Health*, 2(12). [https://doi.org/10.1016/s2589-7500\(20\)30246-6](https://doi.org/10.1016/s2589-7500(20)30246-6)

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